01

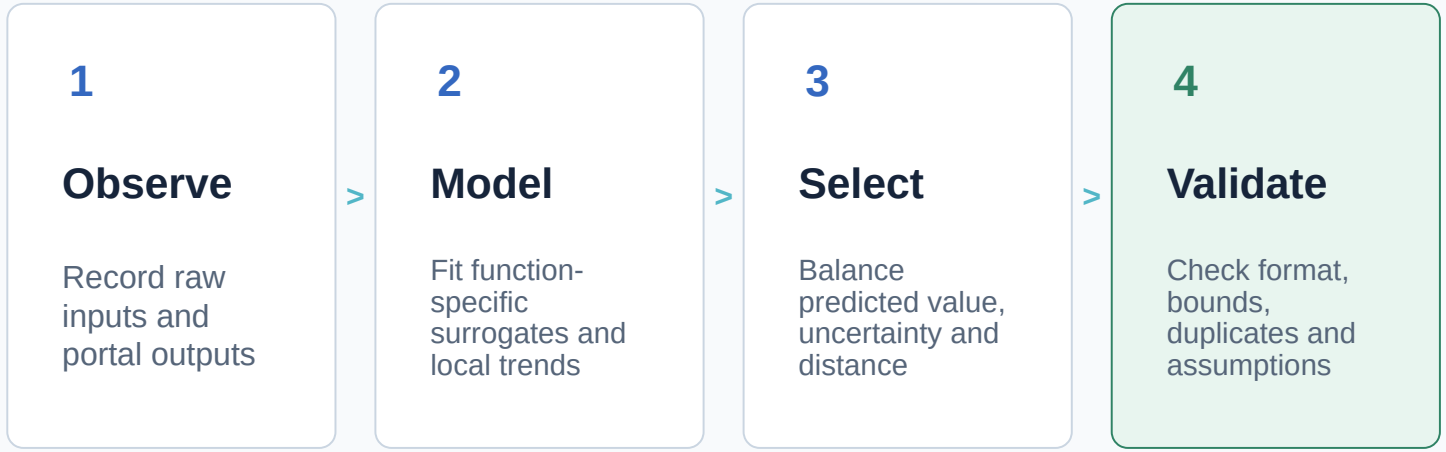
An overview of the BBO approach

Find strong inputs for eight unknown functions with as few expensive evaluations as possible

The objective

Maximise every function while learning from limited feedback. The analytical equations, gradients and global optima remain hidden.

Each query must earn its place: improve the incumbent or reduce uncertainty about the next move.



Current evidence base

263 completed evaluations through Week 11 across 2D to 8D spaces

02

How the strategy evolved

Returned results gradually replaced one general policy with eight local decision rules

Rounds 1–2

Global search

Gaussian Processes, UCB, Expected Improvement and Sobol candidates

Rounds 3–6

Model comparison

Regression, SVM and neural surrogates tested nonlinearity and feature sensitivity

Rounds 7–9

Local adaptation

Function-specific tuning, bracketing and controlled coordinate changes

Rounds 10–12

Transparent structure

Reproducible rules, cluster medoids and PCA-style directional refinement

Current decision rule

Continue a direction only after repeated gains. Otherwise shrink, reverse or isolate one variable.

Main influences: observed trends, failed probes, model disagreement and the need for reproducible reasoning

03

Patterns, data and insights

The strongest signals are local directions and boundary behaviour, not one universal feature rule

Function	Strongest observed pattern	Best output	Best week
1	Compact peak near (0.580, 0.580)	**0.0001362**	**10**
2	Noisy ridge near x2 = 1	**0.685945**	**9**
3	Narrow basin near [0.518, 0.605, 0.424]	**-0.002832**	**6**
4	Stable four-variable direction	**0.678794**	**11**
5	x2 drives gains on a boundary face	**1623.345**	**11**
6	x4 dominates local variation	**-0.305650**	**3**
7	Stable six-variable direction	**1.813268**	**11**
8	Boundary ridge with flattening gains	**9.942963**	**10**

What shapes performance

- Functions 4 and 7 improve along coordinated directions.
- Function 5 reduces locally to one active coordinate.
- Functions 2 and 6 show that small distance does not guarantee similar output.
- Function 8 reached a plateau, so further continuation needs smaller steps.

Result **All eight functions improved on their initial incumbent**

04

Decision-making and iteration

Exploit repeated evidence, preserve small diagnostic probes and adapt when local assumptions fail

What worked

Function 5

A controlled sequence changed only x2 while x1, x3 and x4 stayed on the same boundary face.

Best output

1088.86 initially, 1623.35 by Week 11

What required adaptation

Function 2

Very close inputs returned sharply different values. Smooth interpolation and fixed continuation became unreliable.

Adjustment

Keep x2 at its strong boundary and reduce movement in x1

Exploration remains useful when it distinguishes explanations. Exploitation dominates after a direction improves repeatedly.

05

Next steps and reflection

Use the final rounds to confirm durable directions, protect incumbents and communicate uncertainty clearly

Planned actions

- Continue Functions 4, 5 and 7 only if Week 12 confirms another gain.
- Tighten around incumbents for Functions 1, 3 and 6.
- Use conservative probes for noisy Function 2 and plateauing Function 8.
- Update the query log, datasheet and model card with final observations.

Connection to machine learning

The project mirrors active learning and experimental design: models guide which label to purchase next. It also shows why uncertainty, validation and documentation matter when data are scarce.

Message for a non-technical audience

I tested a small number of carefully chosen settings, learned which regions consistently performed better and narrowed the search. The results improved every function, although they do not prove that the global best setting has been found.

Final principle

Spend the remaining query budget where the evidence is strongest and the uncertainty is decision-relevant.